

Unsolved Problems in Combinatorial Games

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1 Problems

During the recent development of Combinatorial Game Theory, many more problems have been suggested than solved. Here is a collection of problems that many people have found interesting. Included is a discussion of recent developments in these areas. This collection was started by Richard Guy in 1991, and has been updated in each *Games of No Chance* volume. The problems are divided into five sections:

- A. Taking & Breaking
- B. Pushing & Placing Pieces
- C. Playing with Pencil & Paper
- D. Disturbing & Destroying
- E. Theory of Games

In the sections, the problems are identified by letters and numbers. Any number in parentheses is the old number used in each of the lists of unsolved problems given on pp. 183–189 of AMS *Proc. Sympos. Appl. Math.* **43**(1991), called PSAM **43** below; on pp. 475–491 of *Games of No Chance*, hereafter referred to as GONC.

Previous versions of the unsolved problems can also be found on pp. 457–473 of *More Games of No Chance* (MGONC); pp. 475–500 of *Games of No Chance 3* (GONC3); pp. 279–308 of *Games of No Chance 4* (GONC4); and pp. 125–168 of *Games of No Chance 5* (GONC5).

Some problems have little more than the statement of the problem if there is nothing new to be added. References [#] are at the end of this article. Useful references for the rules and an introduction to many of the specific games mentioned below are M. Albert, R. J. Nowakowski & D. Wolfe, *Lessons in Play: An Introduction to the Combinatorial Theory of Games*, AKPeters, 2007 (*LIP*); Berlekamp, Conway & Guy, *Winning Ways for your Mathematical Plays*, vol. 1–4, AKPeters, 2000–2004 (*WW*); and Siegel *Combinatorial Game Theory*, American Math. Society, 2013 (*CGT*).

The term ‘game’ is ambiguous and needs to be clarified. If a board is not specified then the question pertains to all possible boards. Berlekamp noted that many games have a standard opening position, an empty board for example. However, some results include positions, such as Gardens-of-Eden positions, that could never occur by any sequence of moves from a standard opening position, and ones which would never occur in decent play. Indeed, Plambeck has asked at several meetings whether there is any difference in the complexity results for all positions and those that would occur in decent play.

Common usage suggests the term ‘game-name’ as meaning the ruleset without a specified opening board. For example, Drummond-Cole’s *2 in DOMINEERING has ‘holes’ which could not be formed starting from the empty board. The term *Restricted* will be used if the positions are (or must be) obtainable from a standard opening board.

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A. Taking & Breaking Games

A1(1). SUBTRACTION GAMES with finite subtraction sets are known to have periodic nim-sequences. Investigate the relationship between the subtraction set and the length and structure of the period. The same question can be asked about **partizan** subtraction games, in which each player is assigned an individual subtraction set. See Fraenkel & Kotzig [48].

[A move in the game $S(s_1, s_2, s_3, \dots)$ is to take a number of beans from a heap, provided that number is a member of the **subtraction-set**, $\{s_1, s_2, s_3, \dots\}$. Analysis of such a game and of many other heap games is conveniently recorded

by a **nim-sequence**,

$$n_0 n_1 n_2 n_3 \dots,$$

meaning that the nim-value of a heap of h beans is n_h ; i.e., that the value of a heap of h beans in this particular game is the **nimber** $*n_h$.]

For examples see Table 2 in §4 on p. 67 of the Impartial Games paper in GONC.

It would now seem feasible to give the complete analysis for games whose subtraction sets have just three members, though this has so far eluded us. Several people, including Mark Paulhus and Alex Fink, have given a complete analysis for all sets $\{1, b, c\}$ and for sets $\{a, b, c\}$ with $a < b < c < 32$. Mark Daniel Ward has pushed this to $a < b < c < 4096$ and has conjectured [151]: *Case 1:* Suppose $c = a + b$. Let $j \equiv b - a \pmod{2a}$, $0 \leq j < 2a$, then

$$p = \begin{cases} 2b + a - j, & \text{if } 0 \leq j < a \\ a(2b - a + j) / \gcd(a, 2a - j), & \text{if } a \leq j < 2a; \end{cases}$$

Case 2: Suppose $c \neq a + b$ then p is a divisor of at least one of $a + b$, $a + c$ and $b + c$. Moreover, if G is the subset of these terms which p divides then $p = \gcd_{(x+y) \in G} (x + y)$.

In general, period lengths can be surprisingly long, and it has been suggested that they could be super-polynomial in terms of the size of the subtraction set. However, Guy conjectures that they are bounded by polynomials of degree at most $\binom{n}{2}$ in s_n , the largest member of a subtraction set of cardinality n . It would also be of interest to characterize the subtraction sets which yield a purely periodic nim-sequence, i.e., for which there is no preperiod. Carlos Santos [120] reduced this upper bound slightly by using a dynamical system approach.

Angela Siegel [129] (and *LIP*) considered infinite subtraction sets which are the complement of finite ones and showed that the nim-sequences are always arithmetic periodic. That is, the nim-values belong to a finite set of arithmetic progressions with the same common difference. The number of progressions is the period and their common difference is called the **saltus**. For instance, the game $S\{\hat{4}, \hat{9}, \hat{26}, \hat{30}\}$ (in which a player may take any number of beans except 4, 9, 26 or 30) has a pre-period of length 243, period-length 13014 and saltus 4702. Marla Slusky & Danny Sleator have proved her conjecture that the class of games $S\{\hat{a}, \hat{b}, \widehat{a+b}\}$ is purely periodic with period length $3(a+b)$.

For infinite subtraction games in general there are corresponding questions about the length and purity of the period.

Suppose the elements of the finite subtraction sets are not constants. It was shown by Fraenkel [47] that the game with subtraction set $\{1, 2, \dots, t-1, \lfloor n/t \rfloor\}$ where $t \geq 2$ is a fixed integer, n the pile size, has an *aperiodic nim-sequence*. See also Fraenkel [47], Guo [59]. Sopena [132] has a variant, I-MARK(S, D), where S contains the subtraction, and D the division elements so that a move from n is one of $n - s$ for $s \in S$, and $\lfloor \frac{n}{d} \rfloor$ for $d \in D$. He shows that I-MARK($\{1\}, \{2\}$) has an aperiodic nim-sequence but a periodic outcome sequence. The nim-sequence has a ‘period’ except for one element which is non-periodic and the values are taken from $\{0, 1, 2\}$. Larsson & Fox [84] show that the subtraction game with $S = \{F_{2n+1} - 1\}$ has a non-periodic nim-sequence with values from $\{0, 1, 2\}$.

It is possible to get a subtraction-division game with non-periodic nim-sequence with values from $\{0, 1\}$. Write n in binary and a move is to remove a leading 1, but not if n is a power of 2, or remove a trailing 0. The number of moves is fixed so this is a version of she-loves-me-she-loves-me-not. Can this be done with a subtraction game set?

An old, voiced but not written, question asks for: an infinite subtraction set where the associated outcome-sequence is periodic but the nim-sequence is aperiodic. These are easy to find in octal games.

Many n -dimensional subtraction games (subtraction sets of vectors) exhibit a kind of “uniform periodicity.” For example, WHITE KNIGHT (WW, page 57) is a bi-dimensional subtraction game where S is the set of vectors $\{(-2, 1-), (-1, -2), (-2, 1), (1, -2)\}$ and JENGA [19] are very well behaved. However, Larsson has commented that there are cases where such strong periodicity seems not to exist. Is there only a kind of “point-wise periodicity,” i.e., all rows and columns are ultimately periodic, if thought of in isolation? (See E16 for some similar questions in misère.)

A2(2). Are all finite **octal games** ultimately periodic?

[If the binary expansion of the k th code digit in the game with code $\mathbf{d}_0 \cdot \mathbf{d}_1 \mathbf{d}_2 \mathbf{d}_3 \dots$ is

$$\mathbf{d}_k = 2^{a_k} + 2^{b_k} + 2^{c_k} + \dots,$$

where $0 \leq a_k < b_k < c_k < \dots$, then it is legal to remove k beans from a heap, provided that the rest of the heap is left in exactly a_k or b_k or c_k or $\dots$ non-empty heaps. See WW, 81–115. Some specimen games are exhibited in Table 3 of §5 of the Impartial Games paper in GONC.]

Resolve any number of outstanding particular cases, e.g., $\cdot 6$ (OFFICERS), $\cdot 04$, $\cdot 06$, $\cdot 14$, $\cdot 36$, $\cdot 37$, $\cdot 64$, $\cdot 74$, $\cdot 76$, $\cdot 004$, $\cdot 005$, $\cdot 006$, $\cdot 007$, $\cdot 014$, $\cdot 015$, $\cdot 016$, $\cdot 024$, $\cdot 026$, $\cdot 034$, $\cdot 064$, $\cdot 114$, $\cdot 125$, $\cdot 126$, $\cdot 135$, $\cdot 136$, $\cdot 142$, $\cdot 143$, $\cdot 146$, $\cdot 162$, $\cdot 163$, $\cdot 164$, $\cdot 166$, $\cdot 167$, $\cdot 172$, $\cdot 174$, $\cdot 204$, $\cdot 205$, $\cdot 206$, $\cdot 207$, $\cdot 224$, $\cdot 244$, $\cdot 245$, $\cdot 264$, $\cdot 324$, $\cdot 334$, $\cdot 336$, $\cdot 342$, $\cdot 344$, $\cdot 346$, $\cdot 362$, $\cdot 364$, $\cdot 366$, $\cdot 371$, $\cdot 374$, $\cdot 404$, $\cdot 414$, $\cdot 416$, $\cdot 444$, $\cdot 564$, $\cdot 604$, $\cdot 606$, $\cdot 744$, $\cdot 764$, $\cdot 774$, $\cdot 776$ and GRUNDY'S GAME (split a heap into two unequal heaps; *WW*, pp. 96–97, 111–112; *LIP*, p. 142), which has been analyzed, first by Dan Hoey, and later by Achim Flammenkamp, as far as heaps of 2^{35} beans.

J.P. Grossman, GONC5, using a new approach based on the sparse space phenomenon, has analyzed $\cdot 6$ up to heaps of size 2^{47} and has found no periodicity.

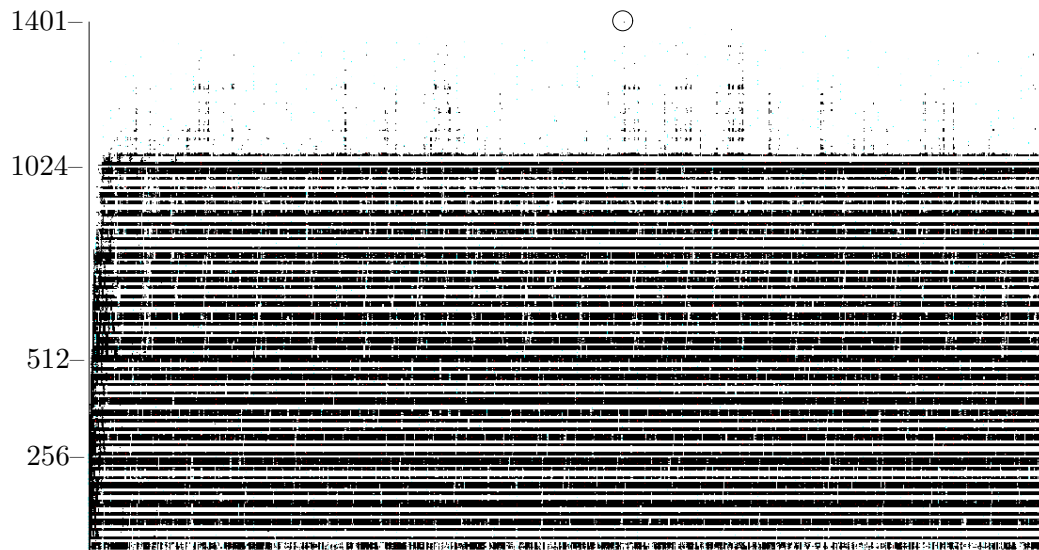


Figure 1: Plot of 11000000 nim-values of the octal game $\cdot 007$

Perhaps the most notorious and deserving of attention is the game **·007**, one-dimensional Tic-Tac-Toe, or TREBLECROSS, which Flammenkamp has pushed to 2^{25} . Figure 1 shows the first 11 million nim-values, a small proportion of which are ≥ 1024 ; the largest, $\mathcal{G}(6193903) = 1401$ is shown circled. Will 2048 ever be reached?

Achim Flammenkamp has settled **·106**: it has the remarkable period and preperiod lengths of 328226140474 and 465384263797. For information on the current status of each of these games, see Flammenkamp’s web page at <http://www.uni-bielefeld.de/~achim/octal.html>

A game similar to Grundy’s, and which is also unsolved, is John Conway’s COUPLES-ARE-FOREVER (*LIP*, p. 142) where a move is to split any heap except a heap of two. The first 50 million nim-values haven’t displayed any periodicity. See Caines et al. [15]. More generally, Bill Pulleyblank suggests looking at splitting games in which you may only split heaps of size $> h$, so that $h = 1$ is She-Loves-Me-She-Loves-Me-Not and $h = 2$ is Couples-Are-Forever. David Singmaster suggested a similar generalization: you may split a heap provided that the resulting two heaps each contain at least k beans: $k = 1$ is the same as $h = 1$, while $k = 2$ is the third cousin of DAWSON’S CHESS.

Explain the structure of the periods of games known to be periodic.

In *Discrete Math.*, **44**(1983) 331–334, Problem 38, Fraenkel raised questions concerning the computational complexity (see **E1** below) of octal games. In Problem 39, he & Kotzig define **partizan octal games** in which distinct octals are assigned to the two players. Mesdal [GONC3, pp. 447–461] show that in many cases, if the game is “all-small” (*WW*, pp. 229–262, *LIP*, pp. 183–207), then the atomic weights are arithmetic periodic. (See **E13** for an explanation of the new name ‘dicotic’ games.) In Problem 40, Fraenkel introduces **poset games**, played on a partially ordered set of heaps, each player in turn selecting a heap and then removing a non-negative number of beans from this heap and from each heap above it in the ordering, at least one heap being reduced in size. For posets of height one, new regularities in the nim-sequence can occur. Tanya Khovanova and the author propose the following simple version. Given n nim-heaps of sizes $a_1, a_2, \dots, a_n$, a move is to choose some heap i and some number $1 \leq x \leq a_i$ and for each $j \geq i$ replace a_j by $\max\{a_j - x, 0\}$. The solution for three heaps involves the continued fraction expansion of $\sqrt{2}$.

Note that this includes, as particular cases, Subset Takeaway, Chomp or Divisors, and Green Hackenbush forests. Compare Problems **A3**, **A13**, **D1** and **D2** below.

A3(3). HEXADECIMAL GAMES have code digits $\mathbf{d}_k$ in the interval from $\mathbf{0}$ to $\mathbf{f}$ ($= \mathbf{15}$), so that there are options splitting a heap into three heaps. See *WW*, 116–117.

Such games may be arithmetically periodic. Nowakowski has calculated the first 100000 nim-values for each of the 1-, 2- and 3-digit games. Richard Austin’s theorem 6.8 in his thesis [4] and the generalization by Howse & Nowakowski [72] suffice to confirm the arithmetic periodicity of several of these games.

Some interesting specimens are $\cdot\mathbf{28} = \cdot\mathbf{29}$, which have period 53 and saltus 16, the only exceptional value being $\mathcal{G}(0) = 0$; $\cdot\mathbf{9c}$, which has period 36, preperiod 28 and saltus 16; and $\cdot\mathbf{f6}$ with period 43 and saltus 32, but its apparent preperiod of 604 and failure to satisfy one of the conditions of the theorem prevent us from verifying the ultimate periodicity. The game $\cdot\mathbf{205200c}$ is arithmetic periodic with pre-period length of 4, period length of 40, saltus 16 except that $40k + 19$ has nim-value 6 and $40k + 39$ has nim-value 14. This regularity, (which also seems to be exhibited by $\cdot\mathbf{660060008}$ with a period length of approximately 300,000), was first reported in Horrocks & Nowakowski [70] (see Problem **A2**.) Grossman & Nowakowski [GONC4] have shown that the nim-sequences for $\cdot\mathbf{200\dots0048}$, with an odd number of zero code digits, exhibit “ruler function” patterns. The game $\cdot\mathbf{9}$ has not so far yielded its complete analysis, but, as far as analyzed (to heaps of size 100000), exhibits a remarkable fractal-like set of nim-values. See Howse & Nowakowski [72]. Also of special interest are $\cdot\mathbf{e}$; $\cdot\mathbf{7f}$ (which has a strong tendency to period 8, saltus 4, but, for $n \leq 100,000$, has 14 exceptional values, the largest being $\mathcal{G}(94156) = 26614$); $\cdot\mathbf{b6}$ (which “looks octal”); $\cdot\mathbf{b33b}$ (where a heap of size n has nim-value n except for 27 heap sizes which appear to be random).

Other unsolved hexadecimal games are $\cdot\mathbf{1x}$ with $\mathbf{x} \in \{8, 9, c, d, e, f\}$;
 $\cdot\mathbf{2x}$, $a \leq \mathbf{x} \leq f$; $\cdot\mathbf{3x}$, $8 \leq \mathbf{x} \leq e$; $\cdot\mathbf{4x}$, $\mathbf{x} \in \{9, b, d, f\}$; $\cdot\mathbf{5x}$, $8 \leq \mathbf{x} \leq f$;
 $\cdot\mathbf{6x}$, $8 \leq \mathbf{x} \leq f$; $\cdot\mathbf{7x}$, $8 \leq \mathbf{x} \leq f$; $\cdot\mathbf{9x}$, $1 \leq \mathbf{x} \leq a$; $\cdot\mathbf{9d}$; $\cdot\mathbf{bx}$, $\mathbf{x} \in \{6, 9, d\}$;
 $\cdot\mathbf{dx}$, $1 \leq \mathbf{x} \leq f$; and $\cdot\mathbf{fx}$ with $\mathbf{x} \in \{4, 6, 7\}$.

A4(53). *N*-heap Wythoff Game. Given $N \geq 2$ heaps of finitely many tokens, whose sizes are $p_1, \dots, p_N$ with $p_1 \leq \dots \leq p_N$. Players take turns removing any positive number of tokens from a *single* heap or removing $(a_1, \dots, a_N)$ from *all* the heaps — a_i from the i -th heap — subject to the conditions: (i) $0 \leq a_i \leq p_i$ for each i , (ii) $\sum_{i=1}^N a_i > 0$, (iii) $a_1 \oplus \dots \oplus a_N = 0$, where $\oplus$ is nim addition. The player making the last move wins and the opponent loses. Note that the classical Wythoff game is the case $N = 2$.

For $N \geq 3$, Fraenkel makes the following conjectures.

Conjecture 1. For every fixed set $K := (A^1, \dots, A^{N-2})$ there exists an integer $m = m(K)$ (i.e, m depends only on K), such that

$$(A^1, \dots, A^{N-2}, A_n^{N-1}, A_n^N), \quad A^{N-2} \leq A_n^{N-1} \leq A_n^N$$

with $A_n^{N-1} < A_{n+1}^{N-1}$ for all $n \geq 1$, is the n th $\mathcal{P}$ -position, and

$$A_n^{N-1} = \text{mex} \left(\{A_i^{N-1}, A_i^N : 0 \leq i < n\} \cup T \right), \quad A_n^N = A_n^{N-1} + n$$

for all $n \geq m$, where $T = T(K)$ is a (small) set of integers.

That is, if you fix $N - 2$ of the heaps, the $\mathcal{P}$ -positions resemble those for the classical Wythoff game. For example, for $N = 3$ and $A^1 = 1$, we have $T = \{2, 17, 22\}$, $m = 23$.

Conjecture 2. For every fixed K there exist integers $a = a(K)$ and $M = M(K)$ such that $A_n^{N-1} = \lfloor n\phi \rfloor + \varepsilon_n + a$ and $A_n^N = A_n^{N-1} + n$ for all $n \geq M$, where $\phi = (1 + \sqrt{5})/2$ is the golden section, and $\varepsilon_n \in \{-1, 0, 1\}$.

In Fraenkel & Krieger [49] the following was shown, inter alia: Let $t \in \mathbb{Z}_{\geq 1}$, $\alpha = (2 - t + \sqrt{t^2 + 4})/2$ ($\alpha = \phi$ for $t = 1$), $T \subset \mathbb{Z}_{\geq 0}$ a finite set, $A_n = (\text{mex} \{A_i, B_i : 0 \leq i < n\} \cup T)$, where $B_n = A_n + nt$. Let $s_n := \lfloor n\alpha \rfloor - A_n$. Then there exist $a \in \mathbb{Z}$ and $m \in \mathbb{Z}_{\geq 1}$, such that for all $n \geq m$, either $s_n = a$, or $s_n = a + \varepsilon_n$, $\varepsilon_n \in \{-1, 0, 1\}$. If $\varepsilon_n \neq 0$, then $\varepsilon_{n-1} = \varepsilon_{n+1} = 0$. Also the general structure of the ε_n was characterized succinctly.

This result was then applied to the N -heap Wythoff game. In particular, for $N = 3$ (so that $K = A^1$) it was proved that $A_n^2 = \text{mex} (\{A_i^2, A_i^3 : 0 \leq i < n\} \cup T)$, where $T =$

$$\{x \geq K : \exists 0 \leq k < K \text{ s.t. } (k, K, x) \text{ is a } P\text{-position}\} \cup \{0, \dots, K - 1\}$$

The following upper bound for A_n^3 was established: $A_n^3 \leq (K + 3)A_n^2 + 2K + 2$. It was also proved that Conjecture 1 implies Conjecture 2.

In Sun & Zeilberger [138], a sufficient condition for the conjectures to hold was given. It was then proved that the conjectures are true for the case $N = 3$, where the first heap has up to 10 tokens. For those 10 cases, the parameter values m, M, a, T were listed in a table.

Sun [137] obtained results similar to those in Fraenkel & Krieger [49], but the proofs are different. It was also proved that Conjecture 1 implies Conjecture 2. A method was given to compute a in terms of certain indexes of the A_i and B_j .

A5(23). BURNING-THE-CANDLE-AT-BOTH-ENDS. Conway and Fraenkel ask us to analyze Nim played with a row of heaps. A move may only be made in the leftmost or in the rightmost heap. When a heap becomes empty, then its neighbor becomes the end heap.

Albert & Nowakowski [2] have determined the outcome classes in impartial and partizan versions (called END-NIM, *LIP*, pp. 210, 263) with finite heaps, and Duffy, Kolpin & Wolfe [GONC3, pp. 419–425] extend the partizan case to infinite ordinal heaps. Wolfe asks for the actual values.

Nowakowski suggested to analyze impartial and partizan END-WYTHOFF: take from either end-pile, or the *same* number from both ends. The impartial game is solved by Fraenkel & Reisner [GONC3, pp. 329–347]. Fraenkel [44] asks a similar question about a generalized Wythoff game: take from either end-pile or take $k > 0$ from one end-pile and $\ell > 0$ from the other, subject to $|k - \ell| < a$, where a is a fixed integer parameter ($a = 1$ is END-WYTHOFF).

There is also HUB-AND-SPOKE NIM, proposed by Fraenkel. One heap is the hub and the others are arranged in rows forming spokes radiating from the hub. Albert notes that this game can be generalized to playing on a forest, i.e., a graph each of whose components is a tree. The most natural variant is that beans may only be taken from a leaf (valence 1) or isolated vertex (valence 0).

The partizan game of RED-BLUE CHERRIES is played on an arbitrary graph, see McCurdy [99]. A player picks an appropriately colored cherry from a vertex of minimum degree, which disappears at the same time. Albert et al.[1] (mistakenly) show that if the graph has a leaf, then the value is an integer. They ask: Is every RED-BLUE CHERRIES position an integer? Herman [65] disproved both by finding a 7-cycle with two leaves that has value $1/2$.

A6(17). Extend the analysis of KOTZIG’S NIM (*WW*, 515–517). Is the game eventually periodic in terms of the length of the circle for every finite move set? Analyze the misère version of KOTZIG’S NIM.

Let $\Gamma(S; n)$ be the outcome of the game where S lists all the moves and n is the size of the circle. Tan & Ward [142] gave more evidence of the periodicity by showing: If $n \in \{1, 3, 5, 7, 15\}$, or if $n \equiv 3 \pmod{5}$ and $n \geq 23$, then $\Gamma(1, 4; n) \in \mathcal{P}$; otherwise, $\Gamma(1, 4; n) \in \mathcal{N}$. They also give several conjectures including: if n is odd and $n \notin \{9, 11, 17\}$ then $\Gamma(3, 5; n) \in \mathcal{P}$ and is in $\mathcal{N}$ otherwise. Their evidence bolsters the periodicity conjecture but also indicates that there is likely to be a lot of noise when n is small.

A7(18). Obtain asymptotic estimates for the proportions of $\mathcal{N}$ -, $\mathcal{O}$ - and

$\mathcal{P}$ -positions in Epstein's PUT-OR-TAKE-A-SQUARE game (*WW*, 518–520).

A8. GALE'S NIM. This is Nim played with four heaps, but the game ends when three of the heaps have vanished, so that there is a single heap left. Brouwer and Guy have independently given a partial analysis, but the situation where the four heaps have distinct sizes greater than 2 is open. An obvious generalization is to play with h heaps and play finishes when k of them have vanished.

A9. EUCLID'S NIM is played with a pair of positive integers, a move being to diminish the larger by any multiple of the smaller. The winner is the player who reduces a number to zero. Analyses have been given by Cole & Davie [25], Spitznagel [134], Lengyel [92], Collins [26], Fraenkel [46] and Nivasch [110]. Lengyel [92] reports that Schwartz first found that (a, b) is the sequential sum of nim-heaps given by the normal continued fraction of a/b , the last number in the continued fraction depends on the presence or absence of fibonacci numbers. Gurvich [61] shows that the nim-value, $g^+(a, b)$ for the pair (a, b) in normal play is the same as the misère nim-value, $g^-(a, b)$ except for $(a, b) = (kF_i, kF_{i+1})$ where $k > 0$ and F_i is the i -th Fibonacci number. In this case, $g^+(kF_i, kF_{i+1}) = 0$ and $g^-(kF_i, kF_{i+1}) = 1$ if i is even and the values are reversed if i is odd. Aaron Siegel notes that this can be restated as: EUCLID is tame, and the fickle positions have the form (kF_i, kF_{i+1}) . The latter have genus 0^1 when i is even and 1^0 when i is odd.

Collins & Lengyel [27] play EUCLID'S NIM with three integers and solve some special cases.

In PARTIZAN EUCLID, from the position (p, q) , where $p = kq + r$, $0 \leq r < q$, Left moves to (q, r) and Right to $(q, (k + 1)q - p)$. The outcome classes are investigated in [100]. Determining the values seems hard. The atomic weights may be easier, see Figure 2 where the *mean atomic-weights* for $0 < q < p \leq 100$ are given, where the coordinates (x, y) are $x = q/p$ and y is mean atomic-weight of (p, q) .

A10(20). D.U.D.E.N.E.Y (*WW*, 521–523) is NIM, but with an upper bound, Y , on the number of beans that may be taken, and with the restriction that a player may not repeat his opponent's last move. If Y is even, the analysis is easy. Some advance in the analysis, when Y is odd, has been made by Marc Wallace, Alex Fink & Kevin Saff.

We can, for example, extend the table of strings of pearls given in *WW*, p.523, with the following values of Y which have the pure periods shown, where $D=Y + 2$, $E=Y + 1$. The first entry corrects an error of $128r + 31$ in *WW*.

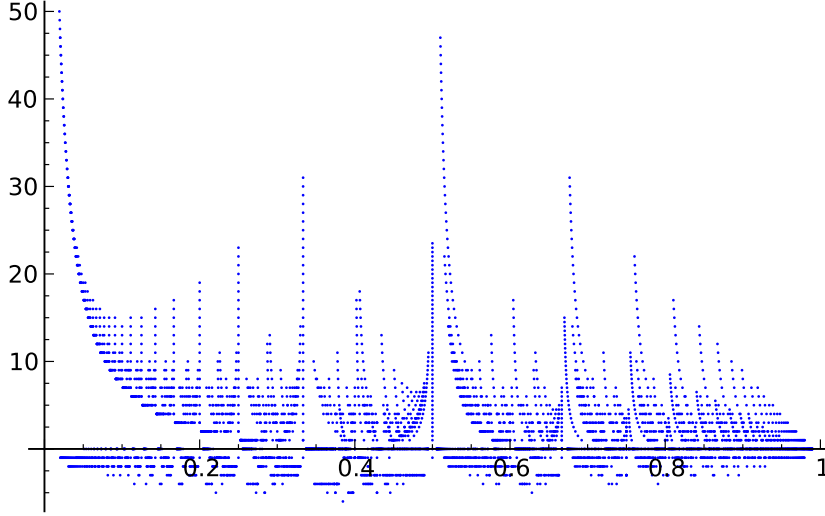


Figure 2: PARTIZAN EUCLID: Mean Atomic Weights for $0 < q < p < 101$

$256r + 31$	DEE	$512r + 153$	DEE	$1024r + 415$	DEE
$512r + 97$	DDEDDDE	$512r + 159$	DEE	$512r + 425$	DE
$1024r + 103$	DE	$512r + 225$	DDE	$512r + 487$	DEE
$128r + 119$	DEE	$512r + 255$	E	$1024r + 521$	DDDE
$1024r + 127$	DEEE	$512r + 257$	DDDDE	$1024r + 607$	DDE
$512r + 151$	DDDEE	$512r + 297$	DDEDEDE	$1024r + 735$	DEEE

It seems likely that the string for $Y = 2^{2k+1} + 2^{2k} - 1$ has the simple period E for all values of k . But the following evidence of the fraction, among 2^k cases, that remain undetermined:

$k =$	3	5	6	7	8	9	10	11	12	13	14	15	16	17
fraction=	$\frac{1}{2}$	$\frac{5}{16}$	$\frac{9}{32}$	$\frac{11}{64}$	$\frac{21}{128}$	$\frac{33}{256}$	$\frac{60}{512}$	$\frac{97}{1024}$	$\frac{177}{2048}$	$\frac{304}{4096}$	$\frac{556}{8192}$	$\frac{974}{16384}$	$\frac{1576}{32768}$	$\frac{2763}{65536}$

suggests that an analysis will never be complete.

Moreover, the periods of the pearl-strings appear to become arbitrarily long.

A11(21). SCHUHSTRINGS is the same as D.U.D.E.N.E.Y, except that a deduction of zero is also allowed, but cannot be immediately repeated (*WW*, 523–524). In *Winning Ways* it was stated that it was not known whether there is any Schuhstring game in which three or more strings terminate simultaneously. Kevin Saff has found three such strings (when the maximum

deduction is $Y = 3430$, the three strings of multiples of 2793, 3059, 3381 terminate simultaneously) and he conjectures that there can be arbitrarily many such simultaneous terminations.

A12(22). Analyze DUDE, i.e., unbounded Dudeney, or Nim in which you are not allowed to repeat your opponent's last move.

Let $[h_1, h_2, \dots, h_k; m]$, $h_i \leq h_{i+1}$, be the game with heaps of size h_1 through h_k , where m is the move just made and $m = 0$ denotes a starting position. Then [28], for $k = 1$ the $\mathcal{P}$ -positions are $[(2s+1)2^{2j}; (2s+1)2^{2j}]$; for $k = 2$ they are $[(2s+1)2^{2j}, (2s+1)2^{2j}; 1]$; and for $k \geq 3$ the heap sizes are arbitrary, the only condition being that the previous move was 1. The nim-values do not seem to show an easily described pattern.

A13. Games with pass moves.

NIM WITH PASS. David Gale suggested an analysis of Nim played with the option of a single pass by either of the players, which may be made at any time up to the penultimate move. It may not be made at the end of the game. Once a player has passed, the game is as in ordinary Nim. The game ends when all heaps have vanished. Morrison, Friedman & Landsberg [106] have looked at this game with their renormalization techniques and suggest that obtaining a full solution may be intractable. Despite that warning, Low & Chan [96] made some progress. They show that:

- (1) for two heaps, $[2m+1, 2m+2]$ are the only $\mathcal{P}$ -positions;
- (2) for three heaps if one is of size 1 then $[1, m, m]$, $m \neq 2$, are the only $\mathcal{P}$ -positions.

They have several other results, typical is: Suppose $p + 2q + 3r \geq 10$ then the position with p heaps of size 1, q heaps of size 2, and r heaps of size 3 is a $\mathcal{P}$ -position iff p is odd, q and r are even, or p is even, q and r are odd.

Chan et al [23] have been extended the analysis to 4 heaps.

CHOCOLATE BAR with a pass [105]. CHOCOLATE BAR is a generalization of CHOMP, see **D1**. Fix a natural number k and a non-negative integer s . For non-negative integers y and z such that $y \leq \lfloor \frac{z+s}{c} \rfloor$ the chocolate bar will consist of $z+1$ columns where the 0th column is the bitter square and the height of the i -th column is $t(i) = \min(y, \lfloor \frac{i+s}{k} \rfloor + 1)$. A move, like CHOMP, is to take a square, remove it and every square in the upper right quadrant based on this square. Let $CB(s, k, y, z, p)$ be the CHOCOLATE BAR game where there is a single pass move which can be used by either player but not at a terminal position. Here $p = 1$ if the pass move is available, $p = 0$ otherwise. For $s < k$,

they solve the game for s odd and ask for a solution when s is even.

In octal games, with a single pass, Horrocks & Nowakowski [70] ask for the regularities that can occur in the $\mathcal{G}$ -sequences. They exhibit sequences which are periodic, arithmetic-periodic and split periodic, arithmetic-periodic.

Now what happens if more than one pass is allowed?

Given a game, the OSLO¹ version gives Left the option to pass in any position except in the terminal positions. Angela Siegel [129] shows that the lattice of OSLO games born by day n is a distributive lattice.

The OSLO version of some games become almost trivial. OSLO WYTHOFF has the values $W(x, y) = 0$ if $x = y = 0$; $W(x, y) = \mathbf{upon}^*$ if $x = y > 0$ or one of x and $y=0$; and $W(x, y) = 2.\mathbf{upon}$ otherwise. (See **E5**, **E7**)

A14. Games with a Muller twist. In such games, each player specifies a condition on the set of options available to her opponent on his next move.

In ODD-OR-EVEN NIM, for example, each player specifies the parity of the opponent's next move. This game was analyzed by Smith & Stănică [131], who propose several other such games which are still open (see also Gavel & Strimling [57]).

The game of BLOCKING NIM proceeds in exactly the same way as ordinary Nim with N heaps, except that before a given player takes his turn, his opponent is allowed to announce a **block**, $(a_1, \dots, a_N)$; i.e., for each pile of counters, he has the option of specifying a positive number of counters which may not be removed from that pile. Flammenkamp, Holshouser & Reiter [69, 68] give the $\mathcal{P}$ -positions for three-heap Blocking Nim with an incomplete block containing only one number, and ask for an analysis of this game with a block on just two heaps, or on all three. There are corresponding questions for games with more than three heaps. Larsson [83] shows with two piles of counters where at most $k - 1$ moves, for some fixed k , may be blocked off at each stage. Then the $\mathcal{P}$ -positions are of the form $\{x, y\}$, where either $|y - x| < k$ and $y - x \equiv k - 1 \pmod{2}$ or $x + y < k$.

Let S be a set of positive integers. The COMPLEMENTARY SUBTRACTION game $\hat{S}$ is played on a heap where the last act of a move is say whether the next subtraction is to be a number from $\hat{S}$ or from its complement. Horrocks & Trenton [71] introduced this variant. They analyzed heaps up to size 8000 the case where S is the Fibonacci numbers without finding periodicity although there appears to be regularity. They also ask about the case $S = \{x | x \equiv$

¹One-Sided-LOopy

$aorb \pmod c\}$. They report that the nim-values appear chaotic. The set of $(a_n, a_n + n)$ where $a_1 = 1$ and $a_n = \text{mex}\{a_i : i < n\}$ form the $\mathcal{P}$ -positions of Wythoff's game. We ask what happens in this game if $S = \{a_n : n+1, 2, 3, \dots\}$?

A15(13). Misère Caternary and Octal games. Misère analysis has been revolutionized by Thane Plambeck & Aaron Siegel [116] with their concept of the **misère quotient** of a game, though the number of unsolved problems continues to increase. See **E13** for more theoretical questions and an explanation of some of the concepts mentioned here.

Plambeck and Siegel ask the specific questions:

(1) The misère quotient of **·07** (DAWSON'S KAYLES) has order 638 at heap size 33. Is it infinite at heap size 34? Even if the misère quotient is infinite at heap 34 then, by Redei's theorem [58, p. 142], [118], it must be isomorphic to a finitely-presented commutative monoid. Call this monoid D_{34} . Exhibit a monoid presentation of D_{34} , and having done that, exhibit D_{35} , D_{36} , etc, and explain what is going on in general. Given a set of games $\mathcal{A}$, describe an algorithm to determine whether the misère quotient of $\mathcal{A}$ is infinite. Much harder: if the quotient is infinite, give an algorithm to compute a presentation for it.

(2) Give complete misère analyses for any of the (normal-play periodic) octal games that show “algebraic-periodicity” in misère play. Some examples are **·54**, **·261**, **·355**, **·357**, **·516** and **·724**. Give a precise definition of algebraic periodicity and describe an algorithm for detecting and generalizing it. This is a huge question: if such an algorithm exists, it would likely instantly give solutions to at least a half-dozen unsolved 2- and 3-digit octals.

Plambeck also offers prizes of US\$500.00 for a complete analysis of DAWSON'S CHESS, **·137** (alias DAWSON'S KAYLES, **·07**); US\$200.00 for the “wild quaternary game”, **·3102**; and US\$25.00 each for **·3122**, **·3123** and **·3312**.

There is a website <http://www.miseregames.org> which contains thousands of misère quotients for octal games.

Siegel notes that Dawson first proposed his problem in 1935, making it perhaps the oldest open problem in combinatorial game theory. [Michael Albert offers the alternative “Is chess a first player win?”] It may be of historical interest to note that Dawson showed the problem to Richard K. Guy around 1947. Fortunately, he forgot that Dawson proposed it as a losing game, was able to analyze the normal play version, rediscover the Sprague-Grundy theory, and get Conway interested in games.

A16 is now **E12**.

A17. MEM and MNEM. MEM is played with heaps of tokens. Remove any number of tokens from any one heap. The number of tokens removed must be at least as large as the number that were removed on the previous move from that heap. Equivalently, a “heap” is a pair of integers (n, k) , and a move is to any pair $(n - i, i)$, where $k \leq i \leq n$.

MNEM is exactly the same as MEM, with additional options: Either player may add tokens to a heap instead of removing them. If adding tokens, the number added must be strictly less than the number of tokens added or removed on the previous move. If removing tokens, the number must be at least as large as the number added or removed on the previous move. So a move from (n, k) is to $(n - i, i)$ for $k \leq i \leq n$, as in MEM, or to $(n + i, i)$ for $1 \leq i < k$.

Conway & (Aaron) Siegel have investigated this game. They conjecture: *Every position in MNEM has finite nim value.* They verified this experimentally up to $n = 10,000$. They also conjecture: *For both MEM and MNEM, if $k^2 \geq n$, then $(n, k) = \lfloor \frac{n}{k} \rfloor$.* Siegel reports that they have no idea how to prove either so these may be very difficult problems. Rubinstein-Salzedo reports that, for MEM, he has a proof.

A18. FIBONACCI NIM. Given $\alpha > 0$, and a heap of n counters. The first move is to take any number except n . On the $m + 1$ st move, $m > 0$, the next player can take up to αq_m , where q_m is the number taken on the m th move. Whinihan [153] attributes this game to R. E. Gaskell, and he solves the case $\alpha = 12$ by showing that the $\mathcal{P}$ -positions are fibonacci numbers. If $\alpha = 1$ then the $\mathcal{P}$ -positions are powers of 2. Schwenk [125] solves the general case using enumeration schemes. Let P_α be the set of $\mathcal{P}$ -positions for a given α . Extending Schwenk’s result, Rubinstein-Salzedo & Sarkar claim that as α increases, then there are intervals on which the $\mathcal{P}_\alpha$ is constant. For example, for $1 \leq \alpha < 2$, $\mathcal{P}_\alpha$ are the powers of 2 together with 0, and for $2 \leq \alpha < 2.5$, $\mathcal{P}_\alpha$ is the set of fibonacci numbers.

Larsson & Rubinstein-Salzedo [91] define the multi-heap game, $\alpha = 2$, in which the first move can remove all of a heap and the restriction on subsequent moves remain regardless of which heap the next player choses. For two heaps, they characterize the $\mathcal{P}$ -positions, again in terms of fibonacci numbers. The cases for different α and for more heaps are open.

See Levine [93] for the game where there is an upper and/or lower bound on the number that can be taken, where the bounds depend upon the size of the heap.

A19. Self-Referential Subtraction Games. There several games under this heading but the best known game under this heading is EUCLID and its variants, see **A9**. A recent partizan variant is SUBVERSION [42], see **E16**. [SUBVERSION is played with a pair of non-negative integers, (a, b) . If $a = 0$ or $b = 0$ then the game is over. Otherwise, if $a \geq b$ then Left can move to $(a, 0)$, if $a < b$ then Left can move to $(a, b - a)$; if $a \leq b$ then Right can move to $(0, b)$, If $a > b$ then Right can move to $(a, b - a)$.]

Two, SUSEN and ALBUS, were developed by Larsson & Nowakowski. In SUSEN² the non-zero heap sizes form the subtraction set. For example, $(2, 3)$ is a $\mathcal{P}$ -position—the moves are to: (2) or (3) both followed by $()$; and to $(2, 1)$ followed by $(1, 1)$. The move to $(2, 2)$ is not available.

If there is only one heap, the next player wins. For two heaps, removing a heap is a losing move. If the players never remove a heap, the length of the game (a, b) until they reach (c, c) for some c , is fixed. Call this $L(a, b)$. Then (a, b) is a $\mathcal{P}$ -positions iff $L(a, b)$ is even. Alex Fink gives a proof that (a, b, c) is in $\mathcal{P}$ iff (a, b) , (a, c) and (b, c) are all in $\mathcal{N}$. Extend the analysis to more heaps.

In ALBUS³, the heap sizes are the illegal moves. For one heap, reduce the heap to 1 counter and win. Two heaps is still unsolved.

B. Pushing & Placing Pieces

B1(5). GO is of particular interest, partly because of the loopiness induced by the “ko” rule, and many problems involve general theory: see **E4** and **E5**.

Elwyn writes:

I attach one region that has been studied intermittently over the past several years. The region occurs in the southeast corner of the board (Figure 3). At move 85 Black takes the ko at L6. What then is the temperature at N4? This position is copied from the game Jiang and Rui played at MSRI in July 2000. In 2001, Bill Spight and I worked out a purported solution by hand, assuming either Black komaster or White komaster.

Elwyn also writes:

Nakamura (GONC3) has shown how capturing races in Go can be analyzed by treating liberties as combinatorial games. Like atomic

²Subtraction Set ENcoded

³ALL BUt set Sizes

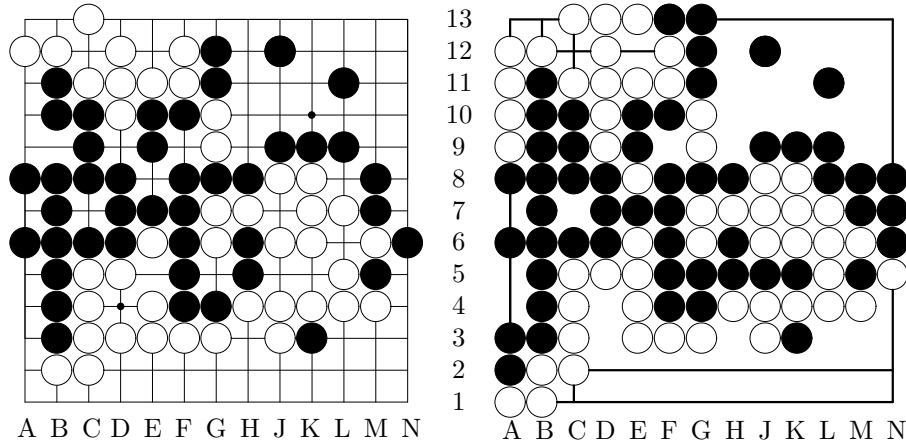


Figure 3: Jiang v. Rui, MSRI, July, 2000

weights, when the values are integers, each player's best move reduces his opponent's resources by one. The similarities between atomic weights and Nakamura's liberties are striking.

Theoretical problem: Either find a common formulation which includes much or all of atomic weight theory and Nakamura's theory of liberties, OR find some significant differences.

Important practical applied problem: Extend Nakamura's theory to include other complications which often arise in GO, such as simple kos, either internal and/or external.

B2. WOODPUSH (see *LIP*, pp. 214, 275) is a game that involves ko but that is simpler than GO. WOODPUSH is played on a finite strip of squares. Each square is empty or occupied by a black or white piece. A piece of the current player's color retreats: Left retreats to the left and Right to the right—to the next empty square, or off the board if there is no empty square; except, if there is a contiguous string containing an opponent's piece then it can move in the opposite direction *pushing* the string ahead of it. Pieces can be moved off the end of the strip. Immediate repetition of a global board positions is not allowed. A "ko" threat must be played first. For example

	Left		Right		Left		Right	
$LRR\Box$	$\rightarrow$	$\Box LRR$	$\rightarrow$	$LR\Box R$	$\rightarrow$	ko-threat	$\rightarrow$	$R\Box\Box R$

Note that Right's first move to $LR\Box R$ is illegal because it repeats the imme-

diately prior board position and Left's second move to $\square LRR$ is also illegal so he must play a ko-threat. Also note that in $\square LRR\square$, Right never has to play a ko-threat since he can always push with either of his two pieces—with Left moving first,

	Left		Right	
$\square LRR\square$	$\rightarrow$	$\square\square LRR$	$\rightarrow$	$\square LR\square R$
	$\rightarrow$	ko-threat	$\rightarrow$	Right answers ko-threat
	$\rightarrow$	$\square\square LRR$	$\rightarrow$	$\square LRR\square$

Berlekamp, Plambeck, Ottaway, Aaron Siegel & Spight (work in progress) use top-down thermography to analyze the three piece positions. What about more pieces? Cazenave & Nowakowski [GONC4, pp. 47–56] show that the position $L.L.R.R$ is ± 4 but that $L.L...R.R$ and $L.L.....R.R$ are draws by superko (repetition of the board position after more than 2 moves).

B3(40). CHESS. Noam Elkies [MGONC pp. 61–78] has examined DAWSON'S CHESS, but played under usual Chess rules, so that capture is not obligatory.

He would still welcome progress with his conjecture that the value $*k$ occurs for all k in (ordinary Chess) pawn endings on sufficiently large chessboards.

Thea van Roode has suggested IMPARTIAL CHESS, in which the players may make moves of either color. Checks need not be responded to and Kings may be captured. The winner could be the first to promote a pawn.

INFINITE CHESS is played with a regular set of chess pieces but on an infinite board. Evans and Hamkins [39] observe that for each finite number n , there is a position in infinite chess, having value n . They show that positions with values ω , ω^2 , $\omega^2 \times k$, and ω^3 occur, and recently, that ω^4 occurs. They ask about 3-dimensional CHESS.

B4(30). CHESS: *King and Rook*. Low & Stamp [97] have given a strategy in which White wins the King and Rook vs. King problem within an 11×9 region. Kanungo & Low [78] show that with the initial position WK(1, 1), WR(x, y) and BK(a, b), where $1 < x < a$, White has a winning strategy on the $(a + b + 3) \times (a + b + 5)$ board.

With the same pieces but on a quarter-infinite board, spanning the first quadrant of the xy -plane, ENTREPENEURIAL CHESS or ECHES, introduced by Berlekamp in 2005, see [8], gives Black is given the additional option of ‘cashing

out’, which removes the board and converts the position into the integer $x + y$, where $[x, y]$ are the coordinates of his king’s position when he decides to cash out. Berlekamp & Low conjecture that the Boiling Point of ECHESS is $5\frac{5}{8}$.

B5. Non-attacking Queens. Noon & Van Brummelen [111] alternately place queens on an $n \times n$ chessboard so that no queen attacks another. The winner is the last queen placer. They give nim-values for boards of sizes $1 \leq n \leq 10$ as 1121312310 and ask for the values of larger boards.

B6(55). AMAZONS. Martin Müller [107] has shown that the 5×5 game is a first player win and asks about the 6×6 game.

B7. PHUTBALL, more properly, Conway’s Philosopher’s Football, is usually played on a Go board with positions (i, j) , $-9 \leq i, j \leq 9$ and the ball starting at $(0, 0)$. For the rules, see *WW*, pp.752–755. The game is loopy (see **E5** below), and Nowakowski, Ottaway & Siegel (see [128]) discovered positions that contained tame cycles, i.e., cycles with only two strings, one each of Left and Right moves. Aaron Siegel asks if there are positions in such combinatorial games which are stoppers but contain a **wild cycle**, i.e., one which contains more than one alternation between Left and Right moves. Sarkar [GONC5] has shown that draws can occur on the 19×19 board but it is not clear if they can occur without collusion of the players. Demaine, Demaine & Eppstein [MGONC, pp. 351–360] show that it is NP-complete to decide if a player can win on the next move. Loosen (see DIRECTIONAL PHUTBALL) raises the question of whether there is any $\mathcal{P}$ -position.

PHLAG PHUTBALL is a variant played on an $n \times n$ board with the initial position of the ball at $(0, 0)$ except that now only the ball may occupy the positions $(2i, 2j)$ with both coordinates even. This eliminates “tackling”, and is an extension of 1-dimensional ODDISH PHUTBALL, analyzed in Grossman & Nowakowski [MGONC, pp. 361–367]. The $(3, 2n + 1)$ board (i.e. (i, j) , $i = 0, 1, 2$ and $-n \leq j \leq n$) is already interesting and requires a different strategy from that appropriate to ODDISH PHUTBALL.

Loosen [95] introduces DIRECTIONAL PHUTBALL, a non-loopy version, which is also played on a grid. The ball starts in the bottom left corner with Left’s goal-line is the right edge, Right’s is the top edge. Players can only jump toward their opponent’s goal-line, and wins by jumping on over that goal-line. Men can only be placed ahead of the ball (i.e. in the positive quadrant with origin on the ball). She notes that there is no $\mathcal{P}$ -position in this game and that making an off-parity move (‘poultry’) can be good. She asks for a complete analysis of the $2 \times n$ board. She shows that the atomic weights of the

$m \times n$ board for $2 \leq m, n, \leq 4$ is 0 and is 0, 1, 1, 2 for $2 \times n$ for $n = 5, 6, 7, 8$ respectively.

B8. HEX. (*LIP*, pp. 264–265) Nash’s strategy stealing argument shows that Hex is a first player win but few quantitative results are known. Arneson et al report that 8×8 is solved, as are most openings on the 9×9 .

Garikai Campbell [16] asks

(1) For each n , what is the shortest path on an $n \times n$ board with which the first player can guarantee a win?

(2) What is the least number of moves in which the first player can guarantee a win? Campbell showed that this is at least n on an $n \times n$ board. Peng et al [115] show that it is 7 on the 5×5 board.

B9(54). FOX AND GEESE. Berlekamp & Siegel [128, Chapter 2] and *WW* pp.669–710, “analysed the game fairly completely, relying in part on results obtained using *CGSuite*.” On p.710 of *WW* the following open problems are given.

1. Define a position’s **span** as the maximum occupied row-rank minus its minimum occupied row-rank. Then quantify and prove an assertion such as the following: If the backfield is sufficiently large, and the span is sufficiently large, and if the separation is sufficiently small, and if the Fox is neither already trapped in a daggered position along the side of the board, nor immediately about to be so trapped, then the Fox can escape and the value is **off**.

2. Show that any formation of three Geese near the centre of a very tall board has a “critical rank” with the following property: If the northern Goose is far above, and the Fox is far below, then the value of the position is either positive, **HOT**, or **off**, according as the northern Goose is closer, equidistant, or further from the critical rank than the Fox.

3. Welton asks what happens if the Fox is empowered to retreat like a Bishop, going back several squares at a time in a straight line? More generally, suppose his straightline retreating moves are confined to some specific set of sizes. Does $\{1,3\}$, which maintains parity, give him more or less advantage than $\{1,2\}$?

4. What happens if the number of Geese and boardwidths are changed?

In Aaron Siegel’s thesis there are several other questions:

5. In the critical position, with Geese at [we use the algebraic Chess notation of a, b, c, d, ... for the files and 1, 2, 3, ..., n for the ranks] (b,n) , (d,n) ,

$(e, n-1)$, $(g, n-1)$, and Fox at $(c, n-1)$, which has value $1 + 2^{-(n-8)}$ on an $n \times 8$ board with $n \geq 8$ in the usual game, is the value $-2n+11$ for all $n \geq 6$ when played with “Ceylonese rules”? (Fox allowed two moves at each turn.)

6. On an $n \times 4$ board with $n \geq 5$ and Geese at (b, n) and $(c, n-1)$ do all Fox positions have value **over**? With the Geese on (b, n) and (d, n) are only other values 0 at $(c, n-1)$ and $\{\mathbf{over}|0\}$ at $(b, n-2)$ and $(d, n-2)$?

7. On an $n \times 6$ board with $n \geq 8$ and Geese at (b, n) , (d, n) and $(e, n-1)$ do the positions $(a, n-2k+1)$, $(c, n-2k+1)$, $(e, n-2k+1)$, all have value 0, and those at $(b, n-2k)$, $(d, n-2k)$, $(f, n-2k)$ all have value $*$? And if the Geese are at (b, n) , (d, n) and (f, n) are the zeroes and $*$ s interchanged?

B10. HARE AND HOUNDS.

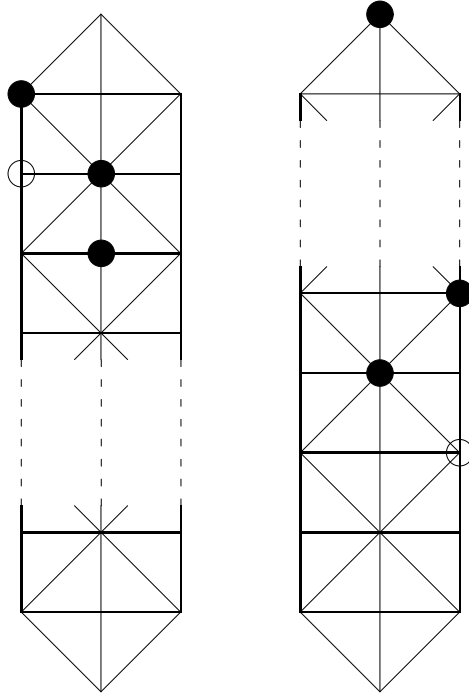


Figure 4: Sequences of Hare and Hounds positions

Aaron Siegel asks if the sequences of positions of increasing board length shown in Figure 4 are, on the left, increasingly hot, and, on the right, have arbitrarily large negative atomic weight. He also conjectures that the starting

position on a $6n+5 \times 3$ board, for $n > 0$, has value

$$-(n-1) + \left\{ b, c \mid 0 \parallel 0 \parallel 0 \parallel 0 \dots \parallel 0 \right\}$$

where there are $2n + 4$ zeroes and slashes and $b = \{0, a \parallel 0, \{0 \mid \mathbf{off}\}\}$,

$$c = \left\{ 0 \parallel \downarrow_{[2]} * \mid 0 \parallel 0 \right\} \text{ and } a = \{0, \downarrow_{[2]} * \mid 0, \downarrow_{[2]} *\}$$

B11(4). DOMINEERING (*WW*, pp. 119–122, 138–142; *LIP* pp. 1–7, 260).
Extend the analysis.

[Left & Right take turns to place dominoes on a checker-board. Left orients her dominoes North-South and Right orients his East-West. Each domino exactly covers two squares of the board and no two dominoes overlap. A player unable to play loses.]

See Berlekamp [5] and the second edition of *WW*, 138–142, where some new values are given. For example David Wolfe & Dan Calistrate have found the values (to within ‘-ish’, i.e., infinitesimally shifted) of 4×8 , 5×6 and 6×6 boards. The value for a 5×7 board is

$$\left\{ \frac{3}{2} \mid \left\{ \frac{5}{4} \mid -\frac{1}{2} \right\}, \left\{ \frac{3}{2} \mid -\frac{1}{2}, \left\{ \frac{3}{2} \mid -1 \right\} \parallel -1 \mid -3 \right\} \parallel -1, \left\{ \frac{3}{2} \mid -\frac{1}{2} \parallel -1 \right\} \mid -3 \right\}$$

Lachmann, Moore & Rapaport [MGONC, pp. 307–315] discover who wins on rectangular, toroidal and cylindrical boards of widths 2, 3, 5 and 7, but do not find their values. Bullock [12, p. 84] showed that 19×4 , 21×4 , 14×6 and 10×8 are wins for Left and that 10×10 is a first player win. Uiterwijk [148] using a new Domineering Solver, shows that Left going first wins on 8×15 , 11×9 , 11×11 , 12×8 , 12×15 , 14×8 and 17×6 boards and loses on 8×12 , 9×11 , and 11×10 . He also shows: Left wins 6×15 , Right wins 11×14 and 11×18 [144]; Left wins 6×19 , 10×15 , 14×15 , 18×15 [145]; and, in [146] proves that (1) all $2k \times n$ boards for $n = 3, 5, 7, 9, 11$ and $2k \times 13$ with $k \geq 3$ are Left wins, and (2) that all $m \times 2k$ boards for $m = 5, 9, 13$, and all $m \times 2k$ boards for $m = 3, 7$ with $k \geq 2$, and all $11 \times 2k$ boards with $k \geq 6$ are Right wins. He also shows [147] how to recursively construct large positions with given values. Drummond-Cole [32] gives outcome classes for many sets of positions. For example, the boards $6 \times n$ is a Left-win for n odd, $n \geq 41$ and $n \neq 43, 45, 51, 59$.

Berlekamp notes that the value of a $2 \times n$ board, for n even, is only known

to within “ish”, and that there are problems on $3 \times n$ and $4 \times n$ boards that are still open.

Berlekamp asks, as a hard problem, to characterize all hot DOMINEERING positions to within “ish”. As a possibly easier problem he asks for a DOMINEERING position with a new temperature, i.e., one not occurring in Table 1 on GONC, p. 477. Also, in particular, is there a DOMINEERING position with temperature greater than 2? In an on-going investigation, Gabriel Drummond-Cole found values with temperatures between 1.5 and 2. Figure 5 shows a position of value $\pm 2^*$ and temperature 2, [31].

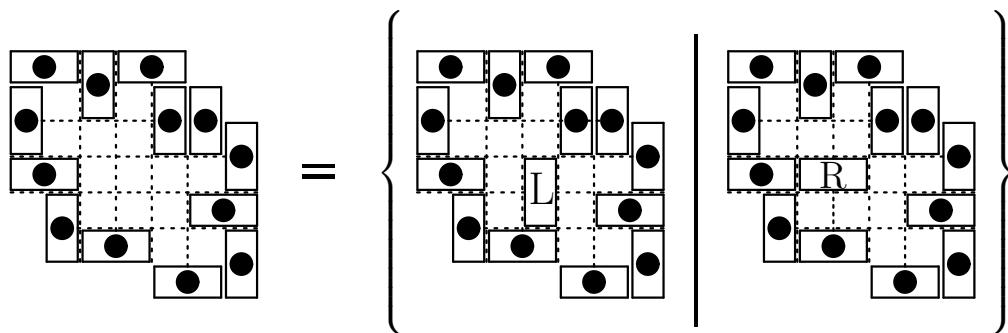


Figure 5: A DOMINEERING position of value $\pm 2^*$

Shankar & Sridharan [126] have found many DOMINEERING positions with temperatures other than those shown in Table 1 on p.477 of GONC. Blanco & Fraenkel [9] have obtained partial results for the game of TROMINEERING, played with trominoes in place of (or, alternatively, in addition to) dominoes.

Dwyer et al [35] found the outcomes of misère DOMINEERING on $2 \times n$ boards

n	0	1	2	3	4	5	6	7	8	9	10	11	≥ 12
$o^-(G_{2 \times n})$	$\mathcal{N}$	$\mathcal{R}$	$\mathcal{P}$	$\mathcal{L}$	$\mathcal{N}$	$\mathcal{R}$	$\mathcal{P}$	$\mathcal{N}$	$\mathcal{N}$	$\mathcal{R}$	$\mathcal{R}$	$\mathcal{N}$	$\mathcal{R}$

B12. NOGo can be found under the name “Anti-Atari Go” at the Sensei Library (see senseis.xmp.net/?AntiAtariGo) and was invented independently by Neil McKay. On a Go-board (or on any graph) pieces are placed as in Go, the only restriction is that every connected group, of each player, must be adjacent to at least one empty intersection. Burke & Hearn [13] have shown that NOGo is NP-hard on a graph. McKay, Nowakowski & (Angela) Seigel found positions of value 1, $1/2$, $1/4$, $1/8$ on a Go board. Using CGSuite, one can find the numbers $*$, $*2$, $*4$, $*8$ on a one-dimensional board. For example,

the positions $x \cdot o \cdot x \cdot o \dots$, where x is a black piece, o a white piece and ‘ $\cdot$ ’ is an empty space, are equivalent to the octal game **.6** (see **A2**) which is not known to be periodic. Not all one-dimensional boards are numbers, for example, $\cdot x \cdot \cdot o \cdot = \pm 1$. We ask for the nim-dimension of both the one- and two-dimensional board. Is there a limit to the denominator of the fractions found on the two-dimensional board?

C. Playing with Pencil & Paper

C1(51). DOTS-AND-BOXES: Sierra [130] reports that 3×3 is a first-player win (see Winning Ways), 4×4 (Wilson) and 4×5 (Barker & Korf) are draws, and that William Fraser reports that 5×5 is a first-player win. See Berlekamp’s book [7] for more problems about this popular children’s (and adults’) game and see also *WW*, pp. 541–584; *LIP*, pp. 21–28, 260.

Several other opening positions have been contemplated. “Open” is the standard position where there are no edges drawn, “Swedish” has all boundary lines drawn, “Faroese” has three, and “Icelandic” has 2—see Figure 6. Jobson et al. [76] show that (1) Faroese $1 \times n$, $n \neq 2$, DOTS-AND-TRIANGLES is a first-player win, and (2) the first player can at least tie on both Open and Faroese DOTS-AND-BOXES $1 \times n$ boards. For the Faroese, they also conjecture that the first Player wins by 1 for all $n \geq 9$ and odd. Sierra has also checked this (using David Wilson’s analyser) for Faroese $1 \times n$, $n \leq 19$ and odd (which he calls staples, S_n), and adds to the conjecture that a winning move for the first player is a vertical move that splits the staple into two staples $S_2 + S_{n-2}$.

Sierra also conjectures for Icelandic DOTS-AND-BOXES with n odd, that the first player loses by 1. Again, this is true for $n \leq 19$.

For misère play, Collette et al [GONC4] show that the Open $1 \times n$ position is a first-player win. They also conjecture: *The outcome of misère $1 \times n$ Swedish Dots-and-Boxes is eventually periodic starting at $n = 22$ and a period of 10.*



Figure 6: Starting position for “Faroese” 1×6 Dots-and-Boxes and Dots-and-Triangles

DOTS-AND-BOXES: Elwyn Berlekamp asks for a complete theory of the

impartial Icelandic $1 \times n$ game. Here, the player who completes the last box loses since they cannot complete their move by drawing another edge.

Larssen et al [89] have extended the normal play theory to cover DOTS & BOXES. However, the analysis of the game is still hard, [76, 29].

C2(25). SPROUTS: Extend the analysis of this Conway-Paterson game in either the normal or misère form. (*WW*, pp. 564–568). Čížek & Balko [24] have developed an AI that plays the normal game perfectly up to 11 dots.

[A move joins two spots, or a spot to itself by a curve which doesn't meet any other spot or previously drawn curve. When a curve is drawn, a new spot must be placed on it. The valence of any spot must not exceed three.]

C3(26). SYLVER COINAGE (*WW*, 575–597): Extend the analysis.

[Players alternately name distinct positive integers, but may not name a number which is a sum, with repetitions allowed, of previously named integers. Whoever names 1 loses.] Sicherman [127] contains the most recent information.

C4(28). Extend Úlehla's or Berlekamp's analysis of VON NEUMANN'S GAME from directed forests to directed acyclic graphs (*WW*, 570–572; Úlehla [150]).

[von Neumann's game, or Hackendot, is played on one or more rooted trees. The roots induce a direction, towards the root, on each edge. A move is to delete a node, together with all nodes on the path to the root, and all edges incident with those nodes. Any remaining subtrees are rooted by the nodes that were adjacent to deleted nodes.]

C5(43). INVERTING HACKENBUSH. Thea van Roode has written a thesis [119] investigating both this and REVERSING HACKENBUSH, but there is plenty of room for further analysis of both games.

In INVERTING HACKENBUSH, when a player deletes an edge from a component, the remainder of the component is replanted with the new root being the pruning point of the deleted edge. In REVERSING HACKENBUSH, the colors of the edges are all changed after each deletion. Both games are hot, in contrast to BLUE-RED HACKENBUSH (*WW*, pp. 1–7; *LIP*, pp. 82, 88, 111–112, 212, 266) which is cold, and GREEN HACKENBUSH (*WW*, pp. 189–196), which is tepid.

C6(42). BEANSTALK and BEANS-DON'T-TALK are games invented respectively by John Isbell and John Conway. See Guy [GONC]. BEANSTALK is played between Jack and the Giant. The Giant chooses a positive integer,

n_0 . Then J. and G. play alternately $n_1, n_2, n_3, \dots$ according to the rule $n_{i+1} = n_i/2$ if n_i is even, $= 3n_i \pm 1$ if n_i is odd; i.e. if n_i is even, there's only one option, while if n_i is odd there are just two. The winner is the person moving to 1.

We still don't know if there are any **$\mathcal{O}$ -positions** (positions of infinite remoteness).

C7(63). The ERDŐS-SZEKERES SEQUENCE GAME [37] (and see Schensted [123]) was introduced by Harary et al [64]. From a deck of cards labelled from 1 through n , Alexander and Bridget alternately choose a card and append it to a sequence of cards. The game ends when there is an ascending subsequence of a cards or a descending subsequence of d cards.

The game appears to have a strong bias towards the first player. Albert et al [GONC3] show that for $d = 2$ and $a \leq n$ the outcome is $\mathcal{N}$ or $\mathcal{P}$ according as n is odd or even, and is $\mathcal{O}$ (drawn) if $n < a$. They conjecture that for $a \geq d \geq 3$ and all sufficiently large n , it is $\mathcal{N}$ with both normal and misère play, and also with normal play when played with the rationals in place of the first n integers.

They also suggest investigating the form of the game in which players take turns naming pairs (i, π_i) subject to the constraint that the chosen values form part of the graph of some permutation of $\{1, 2, \dots, n\}$.

D. Disturbing & Destroying

D1(27). Extend the analysis of CHOMP (*WW*, 598–599, *LIP* 19, 46, 216).

David Gale offered \$300.00 for the solution of the infinite 3-D version where the board is the set of all triples (x, y, z) of non-negative integers, that is, the lattice points in the positive octant of $\mathbb{R}^3$. The problem is to decide whether it is a win for the first or second player.

CHOMP (Gale [55]) is equivalent to DIVISORS (Schuh [124]). CHOMP is easily solved for $2 \times n$ arrays, Sun [136], and indeed a recent result by Byrnes [14] shows that any poset game eventually displays periodic behavior if it has two rows, and a fixed finite number of other elements. See also the Fraenkel poset games mentioned near the end of **A2**.

Thus, most of the work in recent years has been on three-rowed CHOMP. The situation becomes quite complicated when a third row is added, see Zeil-

berger [154] and Brouwer et al. [11]. A novel approach (renormalization) is taken by Friedman & Landsberg (GONC3). They demonstrate that three-rowed CHOMP exhibits certain scaling and self-similarity patterns similar to chaotic systems. Is there a deterministic proof that there is a unique winning move from a $3 \times n$ rectangle? The renormalization approach is based on non-linear dynamics techniques from physics; its results are highly suggestive but as of yet not fully mathematically rigorous.

TRANSFINITE CHOMP has been investigated by Huddleston & Shurman [MGONC]. An open question is to calculate the nim value of the position $\omega \times 4$ —they conjecture this to be $\omega \cdot 2$, but it could be as low as 46, or even uncomputable! Perhaps the most fascinating open question in TRANSFINITE CHOMP is their *Stratification Conjecture*, which states that if the number of elements taken in a move is $< \omega^i$, then the change in the nim-value is also $< \omega^i$.

Consider a poset in which each node is coloured blue, red, or green. Moves consist of removing nodes (blue–Left, red–Right, and green–either) along with all those above them. A solution to the blue–red or to the green poset game would solve a large number of games. The green poset seems very difficult, but the value of a blue–red–poset game is always a number [17]. However, even a general formula for POLYCHROMATIC CHOMP (squares with strawberry and blueberry flavors) has not been found.

Andries Brouwer keeps a website about CHOMP at

<http://www.win.tue.nl/~aeb/games/chomp.html> .

Nakamura & Miyadera [109] introduce a new board. Instead of the columns non-increasing in size going out from the poisoned square, they are non-decreasing. They also add a strip going off to the left. Dozier & Perry [30] introduce more poisoned squares and generalize to ideals, and Grobner bases.

D2(33). SUBSET TAKE-AWAY. Given a finite set, players alternately choose proper subsets subject to the rule that once a subset has been chosen no proper subset can be removed. Last player to move wins.

Many people play the dual, i.e. a non-empty subset must be chosen and no proper superset of this can be chosen. We discuss this version of the game which now can be considered a poset game with the sets ordered by inclusion.

The $(n; k)$ Subset Take-away game is played using all subsets of sizes 1 through k of a n -element set. In the $(n; n)$ game one has the whole set (i.e. the set of size n) as an option, so a strategy-stealing argument shows this must be

a first player win.

Gale & Neyman [56], in their original paper on the game, conjectured that the winning move in the $(n; n)$ game is to remove just the whole set. This is equivalent to the statement that the $(n; n - 1)$ game is a second-player win, which has been verified only for $n \leq 5$. A stronger conjecture states that $(n; k)$ is a second player win if and only if $k + 1$ divides n . This was proved in the original paper only for $k = 1$ or 2 .

Andries Brouwer & J. Daniel Christensen [10] show that these are false. The first player loses in $(7; 7)$ if they take the top set. Instead, they win by taking a set of cardinality 4. The second player wins $(7; 3)$ but 4 does not divide 7.

See also Fraenkel & Scheinerman [53].

D3(39). SOWING or MANCALA games. There appears to have been no advance on the papers mentioned in MGONC, although we feel that this should be a fruitful field of investigation at several different levels.

D4. ANNIHILATION GAMES. k -ANNIHILATION. Initially place tokens on some of the vertices of a finite digraph. Denote by $\rho_{\text{out}}(u)$ the outvalence of a vertex u . A move consists of removing a token from some vertex u , and “complementing” $t := \min(k, \rho_{\text{out}}(u))$ (immediate) followers of u , say $v_1, \dots, v_t$: if there is a token on v_j , remove it; if there is no token there, put one on it. The player making the last move wins. If there is no last move, the outcome is a draw. For $k = 1$, there is an $O(n^6)$ algorithm for deciding whether any given position is in $\mathcal{P}$, $\mathcal{N}$, or $\mathcal{O}$; and for computing an optimal next move in the last 2 cases (Fraenkel & Yesha [54]). Fraenkel asks: Is there a polynomial algorithm for $k > 1$? For an application of k -annihilation games to lexicodes, see Fraenkel & Rahat [52].

D5. TOPPLING DOMINOES (*LIP*, pp. 110–112, 274) is played with a row of vertical dominoes each of which is either blue or red. A player topples one of his/her dominoes to the left or to the right.

See Fink et al [GONC4, pp. 65–76] for the proofs that every number occurs exactly once and is a palindrome; there are exactly n positions with value $*n$. Several conjectures are listed but the most intriguing seems to be: *if G is a palindrome then G ’s value appears uniquely.*

There are several variants of TOPPLING DOMINOES. If all the dominoes must be toppled in the same direction then this is a HACKENBUSH string. TIMBER [114] is an impartial version of TOPPLING DOMINOES played on a

directed graph. The dominoes are on the edges. A player chooses a domino which is toppled in the direction of the edge. The dominoes on incident edges are then toppled (regardless of the underlying edge) and the process is iterated. If only outcomes are required then only trees are interesting. In [114], an algorithm is given to determine the outcome class of a tree, which surprisingly requires nim-values. Values, however, appear to be difficult to determine. The normal and misère versions on a path are related to Dyck paths and Catalan numbers, see **E15** for more details. The partizan version, imaginatively called PARTIZAN TIMBER, has Left and Right dominoes placed on the edges of a directed graph. Any TOPPLING DOMINOES position can be transformed into a PARTIZAN TIMBER position by subdividing each edge and directing the edge toward the new vertex. A similar algorithm to that of TIMBER can be used to determine the outcome class of a tree. The uniqueness of numbers doesn't hold even on a path. In the other direction, are there any values that do not occur in the game?

D6. TOWER OF HANOI. Chappelon et al [22] consider the Tower of Hanoi as a two-player game. The first player can win when there are 3 pegs and it is a draw with 4 or more pegs—parity of moves! They introduce a scoring version (see **E14**) where there are weights on the edges. Almost all are first player wins. Are there any variations with second player wins?

HANOI STICK-UP is played with the disks of the Towers of Hanoi puzzle, starting with each disk in a separate stack. A move is to place one stack on top of another such that the size of the bottom of the first stack is less than the size of the top of the second; the two stacks then fuse (&) into one. The only relevant information about a stack are its top and bottom sizes, and it's often possible to collapse the labelling of positions: so for instance, starting with 8 disks and fusing 1&7 and 2&5

we have stacks	0	1&7	2&5	3	4	6
which can be relabelled	0	1&3	1&2	1	2	3

in which the legal moves are still the same. John Conway, Alex Fink and others have found that the $\mathcal{P}$ -positions of height ≤ 3 in normal Hanoi Stickup are exactly those which, after collapsing, are of the form $0^a 01^b 1^c 12^d 2^e$ with $\min(a + b + c, c + d + e, a + e)$ even, except that if $a + e \leq a + b + c$ and $a + e \leq c + d + e$ then both a and e must be even (02 can't be involved in a legal move so can be dropped).

They also found the normal and misère outcomes of all positions with up to six stacks, but there is more to be discovered.

D7(56). Are there any draws in BEGGAR-MY-NEIGHBOR? Marc Paulhus showed that there are no cycles when using a half-deck of two suits, but the problem for the whole deck (one of Conway’s “anti-Hilbert” problems) is still open. Lakshtanov & Aleksenko [82] showed that the expected number of moves is finite in the modified game where each pile is shuffled when being returned to the deck and that the player to start a new pile is chosen by a Bernoulli trial with $p \in (0, 1)$.

E. Theory of Games

E1(49). Fraenkel updates Berlekamp’s questions on computational complexity as follows:

Demaine, Demaine & Eppstein [MGONC] proved that deciding whether a player can win in a *single* move in PHUTBALL (*WW*, pp. 752–755; *LIP*, p. 212) is NP-complete. Grossman & Nowakowski [MGONC] gave constructive partial strategies for 1-dimensional PHUTBALL. Thus, these papers do not show that PHUTBALL has the required properties.

Perhaps NIMANIA (Fraenkel & Nešetřil [51]) and Multivision (Fraenkel [45]) satisfy the requirements. NIMANIA begins with a single positive integer, but after a while there is a multiset of positive integers on the table. At move k , a copy of an existing integer m is selected, and 1 is subtracted from it. If $m = 1$, the copy is deleted. Otherwise, k copies of $m - 1$ are adjoined to the copy $m - 1$. The player first unable to move loses and the opponent wins. It was proved: (i) The game terminates. (ii) Player I can win. In Fraenkel, Loebl & Nešetřil [50], it was shown that the max number of moves in NIMANIA is an Ackermann function, and the min number satisfies $2^{2^{n-2}} \leq \text{Min}(n) \leq 2^{2^{n-1}}$.

The game is thus intractable simply because of the length of its play. This is a *provable* intractability, much stronger than NP-hardness, which is normally only a *conditional* intractability. One of the requirements for the tractability of a game is that a winner can consummate a win in at most $O(c^n)$ moves, where $c > 1$ is a constant, and n a sufficiently succinct encoding of the input (this much is needed for nim on 2 equal heaps of size n).

To consummate a win in NIMANIA, player I can play randomly most of the time, but near the end of play, a winning strategy is needed, given explicitly. Whether or not this is an “intricate” solution, depends on the beholder. But it seems that it’s of even greater interest to construct a game with a very *simple* strategy which still has high complexity!

Also every play of MULTIVISION terminates, the winner can be determined in linear time, and the winning moves can be computed linearly. But the length of play can be arbitrarily long. So let's ask the following: Is there a game which has

1. simple, playable rules,
2. a simple explicit strategy,
3. length of play at most exponential; and
4. is NP-hard or harder.

Tung [143] proved the following:

Theorem. Given a polynomial $P(x, y) \in \mathbb{Z}[x, y]$, the problem of deciding whether $\forall x \exists y [P(x, y) = 0]$ holds over $\mathbb{Z}_{\geq 0}$, is co-NP-complete.

Define the following game of length 2: player I picks $x \in \mathbb{Z}_{\geq 0}$, player II picks $y \in \mathbb{Z}_{\geq 0}$. Player I wins if $P(x, y) \neq 0$, otherwise player II wins. For winning, player II has only to compute y such that $P(x, y) = 0$, given x , and there are many algorithms for doing so.

Also Jones & Fraenkel [77] produced games, with small length of play, which satisfy these conditions.

So we are led to the following reformulation of Berlekamp's question: Is there a game which has

1. simple, playable rules,
2. a finite set of options at every move,
3. a simple explicit strategy,
4. length of play at most exponential;
5. and is NP-hard or harder.

E2. Complexity closure. Aviezri Fraenkel asks: Are there partizan games G_1, G_2, G_3 such that: (i) $G_1, G_2, G_3, G_1 + G_2, G_2 + G_3$ and all their options have polynomial-time strategies, (ii) $G_1 + G_3$ is NP-hard?

E3. Sums of switch games. David Wolfe considers a sum of games G , each of the form $a||b|c$ or $a|b||c$ where a, b , and c are integers specified in unary.

Is there a polynomial time algorithm to determine who wins in G , or is the problem NP-hard?

E4(52). How does one play **sums** of games with varied overheating operators?

Sentestrat and Top-down thermography (*LIP*, p. 214):

David Wolfe would like to see a formal proof that sentestrat works, an algorithm for top-down thermography, and conditions for which top-down thermography is computationally efficient.

Aaron Siegel asks the following generalized thermography questions.

(1) Show that the Left scaffold of a dogmatic (neutral ko-threat environment; *LIP*, p. 215) thermograph is decreasing as function of t . (Note, this is NOT true for komaster thermographs.) [Dogmatic thermography was invented by Berlekamp & Spight. See [133] for a good introduction.]

(2) Develop the machinery for computing dogmatic thermographs of double kos (multiple alternating 2-cycles joined at a single node).

In the same vein as (2):

(3) Develop a temperature theory that applies to all loopy games.

Siegel thinks that (3) is among the most important open problems in combinatorial game theory. The temperature theory of Go appears radically different from the classical combinatorial theory of loopy games (where infinite plays are draws). It would be a huge step forward if these could be reconciled into a “grand unified temperature theory.” Problem (2) seems to be the obvious next step toward (3).

Conway asks for a natural set of conditions under which the mapping $G \mapsto \int^* G$ is the *unique* homomorphism that annihilates all infinitesimals.

E5. Loopy games (*WW*, pp. 334–377; *LIP*, pp. 213–214) are partizan games that do not satisfy the ending condition. A **stopper** is a game that, when played on its own, has no ultimately alternating, Left and Right, infinite sequence of legal moves. Aaron Siegel reminds us of *WW* 2nd ed. p.369, where the authors tried hard to prove that every loopy game had stoppers, until Clive Bach found the Carousel counterexample. Is there an alternative notion of simplest form that works for *all* finite loopy games, and, in particular, for the Carousel? The simplest form theorem for stoppers is at *WW*, p.351.

Siegel conjectures that, if Q is an arbitrary cycle of Left and Right moves

that contains at least two moves for each player, and is not strictly alternating, then there is a stopper consisting of a single cycle that matches Q , together with various exits to enders, i.e., games which end in a finite, though possibly unbounded, number of moves. [Note that games normally have Left and Right playing alternately, but if the game is a sum, then play in one component can have arbitrary sequences of Left and Right moves, not just alternating ones.]

A long cycle is *tame* if it alternates just once between Left and Right, otherwise it is *wild*. Aaron Siegel writes:

I can produce wild cycles “in the laboratory,” by specifying their game graphs explicitly. So the question is to detect one “in nature”, i.e., in an actual game with (reasonably) playable rules such as PHUTBALL [Problem **B7**].

Siegel also asks under what conditions does a given infinitesimal have a well-defined atomic weight, and asks to specify an algorithm to calculate the atomic weight of an infinitesimal stopper g . The algorithm should succeed whenever the atomic weight is well-defined, i.e., whenever g can be sandwiched between loopfree all-smalls of equal atomic weight.

E6(45). Elwyn Berlekamp asks for the **habitat** of $*2$, where $*2 = \{0, *|0, *\}$.

In DOMINEERING, Uiterwijk & Barton [149] improve on Drummond-Cole’s original $*2$ in that the position is reachable from a standard board (no initial holes) and also exhibit a connected $*3$ position. See, for example, Figure 7, which also shows a Go position, found by Nakamura & Berlekamp [108], whose chilled value is $*2$. The Black and White groups are both connected to life via unshown connections emanating upwards from the second row. Either player can move to $*$ by placing a stone at E2, or to 0 by going to E1. Given a game, let n be the smallest nonnegative integer n such that if a position of the game has value $*k$ then $k < 2^n$.

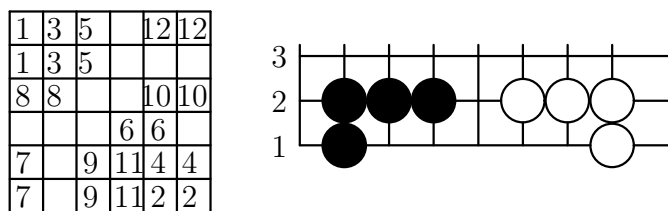


Figure 7: A DOMINEERING position and a Chilled Go position of value $*2$

E7. Partial ordering of games. David Wolfe lets $g(n)$ be the number of games born by day n , notes that an upper bound is given by $g(n+1) \leq g(n) + 2^{g(n)} + 2$, and a lower bound for each $\alpha < 0$ is given by $g(n+1) \geq 2^{g(n)^\alpha}$, for n sufficiently large, and asks us to tighten these bounds. The number born by Day 2 and day 3 are known (see *GCT* p.159). Siegel asks about day 4. Early lower and upper bounds are 3.0×10^{12} and 10^{434} . Suetsugu [140] has improved these to $10^{28.2}$ and 4×10^{184} .

Wolfe also asks what group is generated by the all-small games (or—much harder—of all games) born by day 3. Describe the partial order of games born by day 3, identifying all the largest “hypercubes” (Boolean sublattices) and how they are interconnected. These questions have been answered for day 2 [MGONC, pp. 125–130].

Berlekamp suggests other possible definitions for games born by day n , $\mathcal{G}_n$, depending on how one defines $\mathcal{G}_0$. Our definition is 0-based, as $\mathcal{G}_0 = \{0\}$. Other natural definitions are integer-based (where $\mathcal{G}_0$ are integers) or number-based. These two alternatives do not form a lattice, for if G_1 and G_2 are born by day k , then the games

$$H_n := \left\{ G_1, G_2 \parallel G_1, \{G_2|-n\} \right\}$$

form a decreasing sequence of games born by day $k+2$ exceeding any game $G \geq G_1, G_2$, and the day $k+2$ join of G_1 and G_2 cannot exist. What is the structure of the partial order given by one of these alternative definitions of birthday?

The set of all short games does not form a lattice, but Calistrate, Paulhus & Wolfe [MGONC] have shown that the games born by day n form a distributive lattice $\mathcal{L}_n$ under the usual partial order. They ask for a description of the exact structure of $\mathcal{L}_3$. Siegel describes $\mathcal{L}_4$ as “truly gigantic and exceedingly difficult to penetrate” but suggests that it may be possible to find its dimension and the maximum **longitude**, $\text{long}_4(G)$, of a game in $\mathcal{L}_4$, which he defines as

$$\text{long}_n(G) = \text{rank}_n(G \vee G^\bullet) - \text{rank}_n(G)$$

where $\text{rank}_n(G)$ is the rank of G in $\mathcal{L}_n$ and $G^\bullet$ is the **companion** of G ,

$$G^\bullet = \begin{cases} * & \text{if } G = 0 \\ \{0, (G^L)^\bullet \mid (G^R)^\bullet\} & \text{if } G > 0 \\ \{(G^L)^\bullet \mid 0, (G^R)^\bullet\} & \text{if } G < 0 \\ \{(G^L)^\bullet \mid (G^R)^\bullet\} & \text{if } G \parallel 0 \end{cases}$$

Albert & Nowakowski [3] show that starting with any set of games, instead of just $0 = \{\cdot|\cdot\}$, then the games born on the next day form a complete but not necessarily distributive lattice. They ask which sets of games will give a distributive lattice? Is there a set which will give a non-distributive but modular lattice? Carvalho et al [20, 21] answered this by showing that for any lattice L , there is a set of games which generates L on the next day.

The set of all-small games does not form a lattice, but Siegel forms a lattice $\mathcal{L}_n^0$ by adjoining least and greatest elements $\triangle$ and ∇ and asks: do the elements of $\mathcal{L}_n^0$ have an intrinsic “handedness” that distinguishes, say, $(n-1)\cdot\uparrow$ from $(n-1)\cdot\uparrow + *$?

A game is *option-closed* [113] (now called *hereditarily transitive*) if, recursively, each G^{LL} is also in G^L and the same for Right. For example, HACKENBUSH strings are option-closed. McKay, Nowakowski & (Angela) Siegel [102] show that the option-closed games born on day n form a planar, non-distributive lattice, but the question of *how many?* remains unanswered. Are there other natural families of games that form planar lattices?

E8. Aaron Siegel asks, given a group or monoid, $\mathcal{K}$, of games, to specify a technique for calculating the simplest game in each $\mathcal{K}$ -equivalence class. He notes that some restriction on $\mathcal{K}$ might be needed; for example, $\mathcal{K}$ might be the monoid of games absorbed by a given idempotent.

E9. Siegel also would like to investigate how search methods might be integrated with a canonical-form engine.

E10(9). Develop a **misère theory** for unions of partizan games (*WW*, p. 312).

E11. Four-outcome-games. Guy [62] has given a brute force analysis of a parity subtraction game (start with a subtraction set and an odd number of tokens, the winner is the player who removes an even number of tokens) The Sprague-Grundy theory does not apply because it wasn’t impartial, nor the Conway theory, because it was not last-player-winning. Is there a class of games in which there are four outcomes, $\mathcal{N}$ ext, $\mathcal{P}$ revious, $\mathcal{L}$ eft and $\mathcal{R}$ ight, and for which a general theory can be given? The discussion in Section 2 of [87] hints that there will not be an extension of Normal- or Misère-theory that will solve this type of game.

E12. Impartial Misère Analysis. See **A16**. From the works of Plambeck & Siegel: Let $\mathcal{A}$ be some set of games played under misère rules. Typically, $\mathcal{A}$ is the set of positions that arise in a particular game, such as DAWSON’S

CHESS. Games $H, K \in \mathcal{A}$ are said to be equivalent, denoted by $H \equiv K(\text{mod } \mathcal{A})$, if $H + X$ and $K + X$ have the same outcome for all games $X \in \mathcal{A}$. The relation $\equiv$ is an equivalence relation, and a set of representatives, one from each equivalence class, forms the **misère quotient**, $\mathcal{Q} = \mathcal{A}/\equiv$. A **quotient map** $\Phi : \mathcal{A} \rightarrow \mathcal{Q}$ is defined, for $G \in \mathcal{A}$, by $\Phi : \mathcal{G} = [G]_{\equiv}$.

(1) A quotient map $\Phi : \mathcal{A} \rightarrow \mathcal{Q}$ is said to be **faithful** if, whenever $\Phi(G) = \Phi(H)$, then G and H have the same normal-play Grundy value. Is every quotient map faithful?

(2) Let $(\mathcal{Q}, \mathcal{P})$ be a quotient and $\mathcal{S}$ a maximal subgroup of $\mathcal{Q}$. Must $\mathcal{S} \cap \mathcal{P}$ be nonempty? (Note: it's easy to get a “yes” answer in the special case when $\mathcal{S}$ is the kernel)

(3) Extend the classification of impartial misère quotients. We have preliminary results on the number of quotients of order $n \leq 18$ but believe that this can be pushed far higher.

(4) Exhibit an impartial misère quotient with a period-5 element. Same question for period 8, etc. We've detected quotients with elements of periods 1, 2, 3, 4, 6, and infinity, and we conjecture that there is no restriction on the periods of quotient elements.

(5) In the flavor of both (3) and (4): What is the smallest quotient containing a period 4 (or 3 or 6) element?

E13. Inverses in Partizan misère games. Siegel [Misère Canonical Forms of Partizan Games, GONC4] and Allen [GONC4] extends the misère moniod concepts to partizan games. In normal play, games in both players have a move or neither does are called *all-small* because their values are infinitesimal. This is not true in misère play and the term *dicotic* has been coined to refer to these games in all play conventions. (See **E14** as well.)

Recall that $-G$ is obtained from the game G by reversing the roles of Left and Right—‘turning the board around’. In normal play, $G - G = 0$ which is decidedly not true in misère play. To avoid confusion, when dealing with misère play $-G$ is frequently written as $\overline{G}$ and is called the *conjugate* of G .

(1) Allen asks, in the dicotic universe, $\mathcal{D}$, when is it true that $G + \overline{G} \equiv 0(\text{mod } \mathcal{D})$? This was shown to hold for all games in [85].

(2) Rebecca Milley asks, in a universe, $\mathcal{U}$, closed under sums, conjugation and sub-positions if $G + H \equiv 0(\text{mod } \mathcal{U})$ is it necessarily true that $H = \overline{G}$? Jason Brown had asked the question about arbitrary universes. Milley [103] has the counterexample: on a finite strip of squares, Left places a 1×1 piece

and Right places a 1×2 (domino). In the universe of sums of strips a strip of length 1 plus a strip of length 2 is equivalent to 0. This leads to a different question (see Milley & Renault GONC5): in what universes $\mathcal{U}$ do we have no non-conjugate inverses?

Fisher et al [43] and Milley & Renault [101] respectively characterize the invertible dicot and dead-ending games

A universe $\mathcal{U}$ is *reverse-friendly* if for every $G = \{A \mid \{\cdot \mid B\}\}$, there is $H = \{A' \mid \{\cdot \mid B\}\}$ where H is invertible modulo $\mathcal{U}$ and $G \geq \{\cdot \mid B\} \geq \{\cdot \mid C\}$. Adapting the proofs from [87], it appears that non-conjugate inverses cannot occur in any parental, dense, and reverse-friendly universes.

Santos asks for what universes does the pocancellative property hold: that is, If $G + X \geq H + X$ then $G \geq H$. McKay, Milley, Nowakowski and Santos claim that dead-ending universe is pocancellative. They identify a new universe (an extension of dead-ending), for now labelled $\mathcal{F}$, that has the extended property: if $G, H \in \mathcal{F}$ and X is any misère game then $G + X \geq H + X$ implies $G \geq H$.

Also, they ask if dicots, dead-ending and full misère are cancellative, i.e., if $G + X = H + X$ implies $G = H$. For impartial, this is an old Conway's question. And it was proved by Aaron Siegel. Observe that pocancellative implies cancellative, but the other direction is not true (it is possible to have cancellative and not pocancellative)

E14. Scoring Games. Instead of ‘the last play determines the winner’, another natural way is to have ‘whichever player has the higher score wins’. Both DOTS-AND-BOXES and Go, for example, are scoring games.

Milnor looked at dicotic scoring games with nonnegative incentives. Milnor [104] defines a “positional game” to be what we would call a dicotic (**E13**) scoring game, but only looks at positional games with nonnegative incentives. Hanner [63] looked at the same class of games as Milnor, and invented thermography. Many scoring graph games are covered by Milnor's work See Duchene et al [34] and the Scoring Survey [88]

Ettinger [38] looked at dicotic scoring games (possibly with negative incentives), which he calls ‘positional games’ following Milnor. Johnson [75] looked at the dicotic scoring games that are ‘well-tempered’ in some sense. (A subset of Ettinger's games). Stewart [135] looks at general scoring games. Since the games are no longer dicotic, a choice has to be made about how to handle the situation where the current player can't move but their opponent can.

Larsson, Nowakowski & Santos [86, 85] have developed the theory of Guaranteed Scoring Games. At the end of the game, the score is indicated by $\emptyset^a$ —the empty set to indicate there is no move and a is the score. A Left atomic game has the form $\langle \emptyset^\ell \mid A, B, \dots \rangle$ —Left has no move and $A, B, \dots$ are the Right options. In a Right atomic game, it is Right who has no move. A scoring game G is guaranteed if for every Left atomic game $\langle \emptyset^\ell \mid A, B, \dots \rangle$ in G , if $\emptyset^r$ occurs in one of the Right options $A, B, \dots$ then $\ell \leq r$ and the reverse needs to be true for a Right atomic game. Normal play games can be embedded in Scoring Games by replacing every occurrence of $\emptyset$ by $\emptyset^0$. This mapping is an order-embedding into the set of Guaranteed Scoring Games. They ask: Is the set of Guaranteed Scoring games the largest subset of Scoring games (closed under disjunctive sum and conjugation) in which normal play games can be order-embedded? This subset of Scoring games cannot contain games of the form $\langle \emptyset^\ell \mid \emptyset^r \rangle$ where $\ell > r$.

They ask for a solution of BRUSHES, see the Scoring Survey [88] for the rules.

E15. Find a formula for the number of $\mathcal{P}$ -positions for NIM played with $2n$ tokens. there are none with an odd number of tokens. The On-line Encyclopedia of Integer Sequences (OEIS) has the value for small values of n . Khovanova & Xiong [80] have formulas for when (i) the number of heaps is restricted and (ii) when the number in a heap is restricted.

In general, given a game, enumerate the number of $\mathcal{P}$ -positions of a given size. We only know of TIMBER (D5) where the number of $\mathcal{P}$ -positions, both normal and misère are related to Dyck paths with certain properties (also to Catalan numbers and Fine numbers) and Heteyi [66, 67] which relates the number of $\mathcal{P}$ -positions to the Bernoulli numbers of the second kind.

E16. Miller & Guo (see [60] for an overview) consider games played on lattice points in a polyhedra (often a Cartesian grid in 2 or more dimensions). There are finitely many terminal points, and finitely many vectors which indicate the moves. They originally conjectured that all lattice games have a strategies that can be computed efficiently. Fink [41] disproved this. In *Square-free* lattice game, the vectors only have 0s and 1s as entries. These generalize octal, hexadecimal and many other heap games. Miller & Guo conjecture [60] that there are efficient algorithms for determining the winning moves from any $\mathcal{N}$ -position.

E17. A game has *nim dimension* n if it contains a position $*2^{n-1}$ but not

$*2^n$ (Santos [121]). A game has infinite nim dimension if all the numbers can be constructed. It has null, or $\emptyset$, nim dimension if $*$ cannot be constructed. This was motivated by **E6**.

One can ask for the nim dimension for each game one encounters. TOPPLING DOMINOES (**C5**), THINNING THICKETS ([73]), and KONANE [122] all have infinite nim dimension, COL has nim dimension 1 (WW), It is conjectured that: DOMINEERING (**B11**, **E6**) has nim-dimension = 1; CLOBBER has nim dimension = 1; and AMAZONS **B6** has nim dimension = 3 (only $*4$ is known). Berlekamp asked about NIMSTRING, Santos conjectures it has infinite nim dimension.

E18. *Atomic Weight Calculus and nim-dimension.*

SUBVERSION (also see **A19**.) is a dicotic game in which both players have exactly one move and, in all positions, at least one of the players can end the game. The nim-dimension is 0 since the only numbers that can occur are 0 and $*$. Fisher et al, [42], show that the exceptional case in the atomic weight calculations involve only a finite number of cases. They ask if this is true for any game with bounded nim-dimension. Of course, if true then the number of cases will probably increase with the nim-dimension.

E19. *Universal Partizan Games.* NIM is *universal* for impartial games in that for every impartial value there is (at least) one NIM of that value. Carvalho & Santos show that Generalized KONANE (GONC5) is universal for the Short Conway's Group. Suetsugu has shown that TURNING TILES is a universal ruleset. In [141], he presents two new universal rulesets as well as a universal dicotic ruleset. Are there any other universal games? (Other than sliding a coin down a game-tree.) TOPPLING DOMINOES (**D5**) is a candidate but dealing with reversible options has proven difficult.

In a related question, what is the smallest board on which a set of values can be constructed. The boards in generalized KONANE become huge when constructing the values of Day n . However, all Day 2 values can be found using at most 15 squares of an 8×8 board. Abuku and Suetsugu report that fewer squares are required when playing HEY, THAT'S MY FISH.

A related question: are there any general techniques that, given a game, show that certain values do not occur?

E20. *Temperature.* Nowakowski & Santos define the *boiling point* of a rule set R as $BP(R) = \limsup_G temp(G)$. (See **B11**, **E17**.)

As with nim dimension, one can ask for the boiling point for each game

one encounters. TOPPLING DOMINOES (**D5**), THINNING THICKETS ([73]), and Generalized KONANE ([18]) all have infinite boiling point. It is conjectured that the boiling point of DOMINEERING (**B11**, **E6**) is 2. See Berlekamp [6].

Find feasible methods that give good upper bounds on the temperatures. Nowakowski & Santos show that if the length of the confusion intervals are bounded by k then the boiling point is no more than $\frac{3k}{2}$.

E21. *Joins of games.* When playing game A a position may be a valid position for another game B .

This has been utilized in *conjoined games*, see Huggan, Nowakowski [74]. Start playing A and when a terminal position is reached, start playing B . The two games must be compatible in some not-well-defined sense. This is a possible mechanism to generate a starting position for games without a prescribed start.

Duchêne et al [33] makes it an option for a player to switch games (from A to B) at any point. Tennenhouse makes it an option to switch from normal to misère (or the reverse depending on the rules of the original game). Conway (*WW* p.534) introduced WHIM that is NIM with the extra option, which can only be exercised once, of deciding if the outcome convention is normal or misère.

Farr & Ho [40] explore another possible join.

E22. *Multi-player games* (Propp [117] and [81] are good places to start.) If a player cannot win but can make either of the other two the winner, how should play proceed?

One way is to invoke a preference. Nowakowski, Santos & Silva consider the podium rule (each player wants to be highest on the podium). They consider THREE PLAYER NIM and determine the equivalence classes of the associated monoid. Suetsugu [139] gives each player a preference ordering. An intriguing aspect is that a player doesn't necessarily want themselves to win.

Using the podium rule, misère THREE PLAYER NIM with one pile might be easier to analysis. Both Kelly [79] and Zhao & Liu [155] consider this game. Liu & Duan [94] finally solves the one heap and also the case when the number of players is at least as large as the number of heaps. The other case is still open. They ask for an analysis under other preferences.

We can consider cooperative two- or more player games such that all players want that one of the players make the last move. For two-player games this is a 'puzzle' rather than a game. However, disjunctive sums still occur and the understanding about the recursive construction, comparison procedures, and

canonical forms is an open field.

E23. *Games with Entailing and Complimenting moves.* In *WW*, there are two games, STRINGS-AND-COINS (see **C1**), and TOP ENTAILS, that violate the alternating move condition, thereby disabling a disjunctive sum approach. The first has a complimenting/carry-on move, and the second has an entailing move (that restricts the options) that violate the alternating move condition. They note that both games have *loony* positions. Larsson, Nowakowski, & Santos [89] extend the theory of normal play impartial games (using global threats) to include games that have these types of moves. At the first MSRI Combinatorial Games Conference, Conway proposed finding a ruleset with either of these moves that had a complete solution. The first such game is in [90]. Are there any other natural games?

The theory of partizan entailing games is being prepared. See [36, 98] for some work on CHESS problems, and [152] for some partial results on TOP ENTAILS.

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